

Technical Roadmap of this dummy project

Abstract:-

The objective of this pilot is to validate the core architecture of ScholarSync using a small academic library (10–15 ebooks) while preserving the same workflow and system design that will later scale to millions of books.

This pilot is intended to prove:

- Knowledge ingestion
- Retrieval-Augmented Generation (RAG)
- Citation-backed content generation
- AI authoring workflow
- Basic manuscript review capabilities
- Export and user interaction layer

The pilot will focus on architecture validation rather than production-scale throughput.

Phase 1: Knowledge Base Construction

Scope

Create a small academic knowledge repository using 10–15 ebooks.

Tasks

- PDF/Ebook ingestion
- Text extraction and cleaning
- Chapter and section identification
- Metadata extraction
- Page mapping preservation
- Semantic chunk creation

Deliverables

- Structured document corpus

- Chunked knowledge repository
 - Metadata index
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Phase 2: Embedding & Vector Infrastructure

Scope

Convert processed content into searchable semantic knowledge.

Tasks

- Generate embeddings for document chunks
- Store vectors with metadata
- Maintain source references
- Configure retrieval pipeline

Technology

- Embedding Model (OpenAI / BGE / Instructor)
- Vector Database (ChromaDB / Qdrant / FAISS)

Deliverables

- Searchable vector database
 - Metadata-linked retrieval system
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Phase 3: Citation-Aware Retrieval Engine

Scope

Ensure every generated statement can be traced back to source material.

Tasks

- Semantic search implementation
- Top-K retrieval
- Source ranking
- Citation tracking

- Page reference mapping

Output

Retrieved Context:

- Book Name
- Chapter
- Page Number
- Source Passage

Deliverables

- Citation-aware retrieval layer
-

Phase 4: Author Agent

Scope

Create an AI agent capable of generating academic content from retrieved sources.

Workflow

Input:

- Topic
- Syllabus Point
- Prompt

Process:

- Retrieve relevant context
- Generate chapter outline
- Expand outline into content
- Attach citations

Output:

- Structured academic draft

Deliverables

- Author Agent MVP
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Phase 5: Research Paper Generator (Basic)

Scope

Demonstrate prompt-to-paper generation workflow.

Tasks

Generate:

- Title
- Abstract
- Introduction
- Literature Review
- Conclusion
- References

Notes

This module will use retrieved knowledge from the pilot library and will serve as a proof-of-concept for future automated publication workflows.

Deliverables

- Research Paper Generation Prototype
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Phase 6: Academic Review Agent

Scope

Provide document quality analysis for uploaded content.

Features

Grammar Review

- Grammar correction
- Spelling correction

- Writing quality suggestions

Structure Review

- Heading validation
- Section consistency
- Formatting recommendations

Citation Review

- Missing citation detection
- Citation consistency checks

Deliverables

- Academic Review Assistant
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Phase 7: Originality & Similarity Analysis

Scope

Provide basic plagiarism and duplication analysis.

Features

- Similarity detection against pilot library
- Duplicate content identification
- Rewrite suggestions
- Originality scoring

Deliverables

- Basic originality checker
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Phase 8: Export Layer

Scope

Generate publication-ready outputs.

Supported Formats

- PDF
- DOCX

Features

- Chapter formatting
- Table of contents
- Citation formatting
- Bibliography generation

Deliverables

- Export service
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Phase 9: User Interface

Scope

Create a lightweight web interface for demonstration.

Features

Content Generation

User enters:

- Topic
- Syllabus
- Prompt

System returns:

- Outline
- Generated Content
- Citations

Research Paper Generation

User enters:

- Research Topic

System returns:

- Draft Paper

Document Review

User uploads:

- PDF
- DOCX

System returns:

- Grammar Feedback
- Structure Feedback
- Similarity Analysis

Deliverables

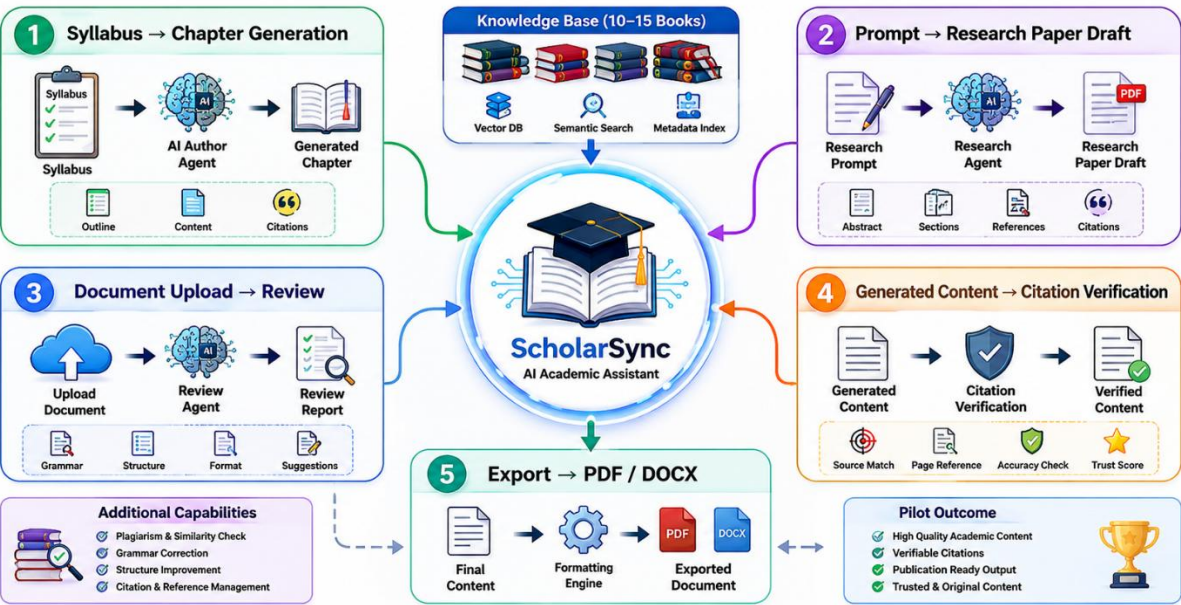
- Interactive Demo Portal

Phase 10: End-to-End Demonstration:-

Demo how's it work-

ScholarSync Pilot – 5 Demo Scenarios

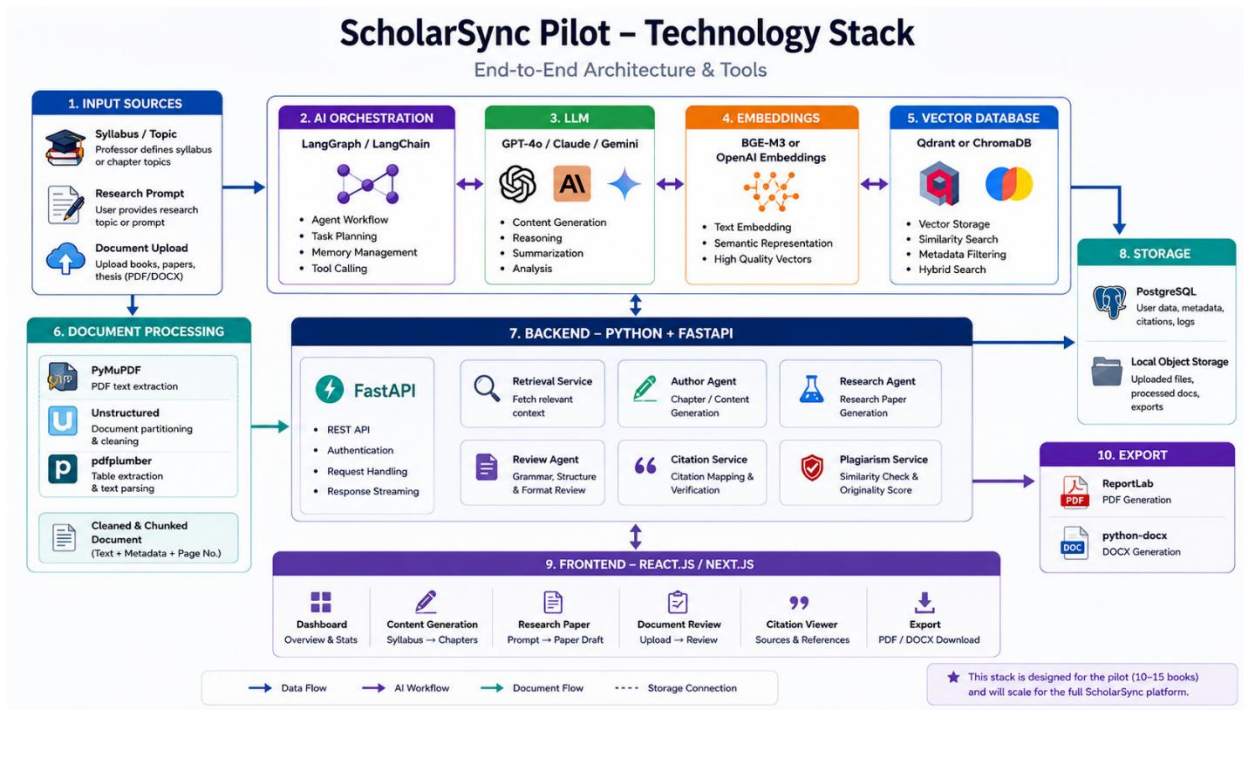
AI Powered Academic Content & Publication Platform



SCHOLARSYNC PILOT – 5 DEMO SCENARIOS



Proposed Pilot Tech Stack



Out of Scope for Pilot

The following features belong to the production-scale version and are intentionally excluded:

- 2M+ ebook ingestion
- Distributed vector clusters
- Multi-agent swarm architecture
- Real-time large-scale generation
- 10,000 manuscripts/day throughput
- AI audio generation
- Concept map generation
- Enterprise monitoring & analytics
- Multi-region deployment

Expected Outcome

At the completion of the pilot, the client will have a working proof-of-concept capable of:

1. Ingesting academic books.
2. Retrieving relevant knowledge.
3. Generating citation-backed academic content.
4. Producing basic research paper drafts.
5. Reviewing uploaded manuscripts.
6. Performing originality analysis.
7. Exporting publication-ready documents.

The pilot will validate the core architecture and provide the foundation for future scaling into the full ScholarSync platform.